Considerations for AI Tools in the Classroom

Given the rapid pace of technological innovation and development, higher education, like nearly all industries, is continuously called upon to consider creative approaches to teaching and learning. The following resource offers instructors a brief introduction to Artificial Intelligence (AI) Tools, specifically ChatGPT, along with several strategies they might consider for navigating or engaging with these tools in their courses.

On this page:

For instructors: share your thoughts with us!

Have additional questions about AI Tools in your classroom? Have examples of how you are integrating these tools in your course or redesigning your assignments to address them? Email CTLFaculty@columbia.edu to share your questions and innovations!

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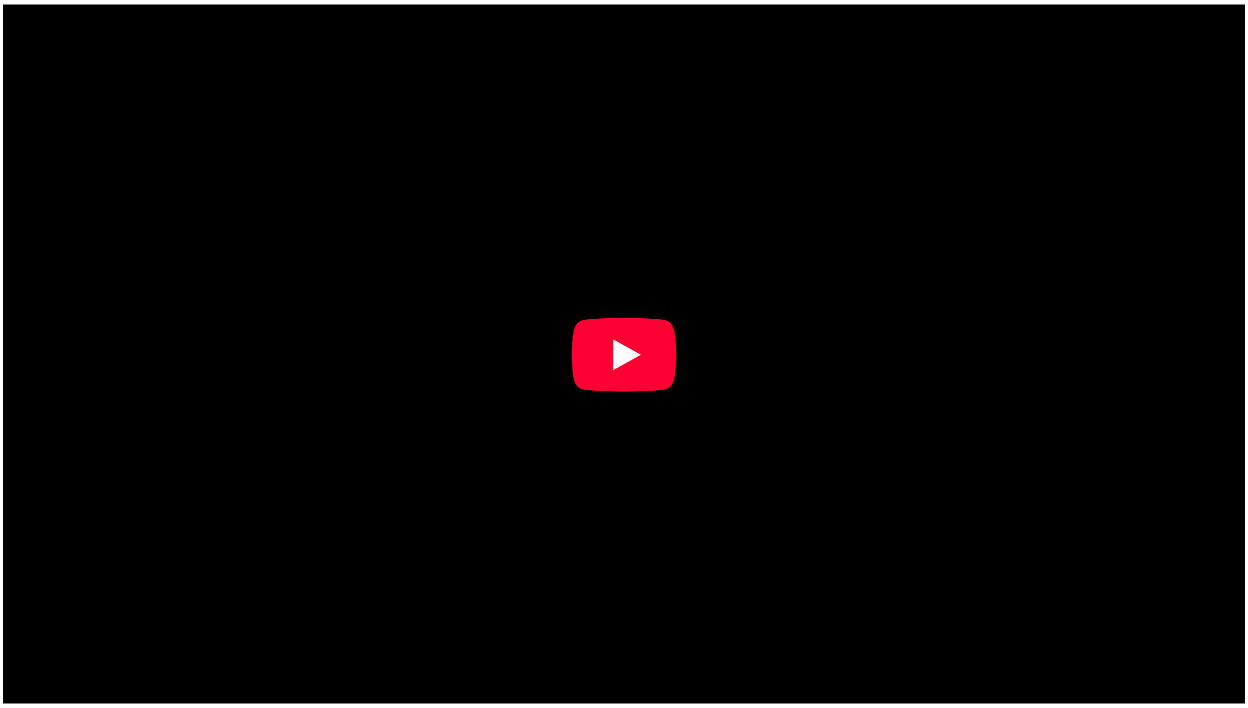
What are AI Tools?

While instructors will continue to encounter new tools and technological innovations, which can sometimes feel overwhelming, the science of teaching and learning offers support for navigating and responding to innovative tools, expanded opportunities, and ongoing shifts.

Most recently, higher education has grappled with the seemingly-overnight introduction of ChatGPT, which first became available in November 2022 and has since captivated the discourse in higher education with its ongoing evolution and iterations. The following section offers an overview of ChatGPT, what it does, and its current limitations. It's important to recognize, however, that because the technology is self-learning, it can improve and evolve quickly; thus, what is a limitation today could be addressed in future iterations or versions. This was perhaps made most evident with the release of GPT-4 during the Spring 2023 semester, which introduced a series of new capabilities for the generative AI tool. Furthermore, since the release of ChatGPT hundreds of AI apps have exploded into the scene, some of which assist in research (e.g. consensus.app and scite.ai), brainstorming (e.g. mymap.ai), and student presentations (e.g. lenovo.ai, twelvelabs.io, and invideo.io).

ChatGPT

ChatGPT, created by OpenAI, is a "model [...] which interacts in a conversational way" and has the ability to "answer followup [sic] questions, admit its mistakes, challenge incorrect premises, and reject inappropriate requests" (OpenAI Blog,"ChatGPT"). In respect to the classroom, ChatGPT can produce written responses to input prompts, write essays and poems, assist with computer code, provide feedback on student-written text, and more.



View a faculty spotlight of Robert J. Morais, Lecturer in Business, and Kamel Jedidi, Jerome A. Chazen Professor of Global Business, who share their experience using ChatGPT in their course.

The most recent version, GPT-4, is described as "more creative and collaborative" with the ability to "generate, edit, and iterate with users on creative and technical writing tasks" (OpenAI, GPT-4). Unlike its predecessors, GPT-4 is capable of producing longer form responses, with more complex analyses in response to particular prompts. It has wider capabilities such as image analysis, and the addition of plug-ins that connect ChatGPT to other third party services, including information from the web. It is important to note, however, that GPT-4 and its capabilities are only available through a *paid subscription* to GPT Plus; these more advanced features are not built into the free uses of ChatGPT.

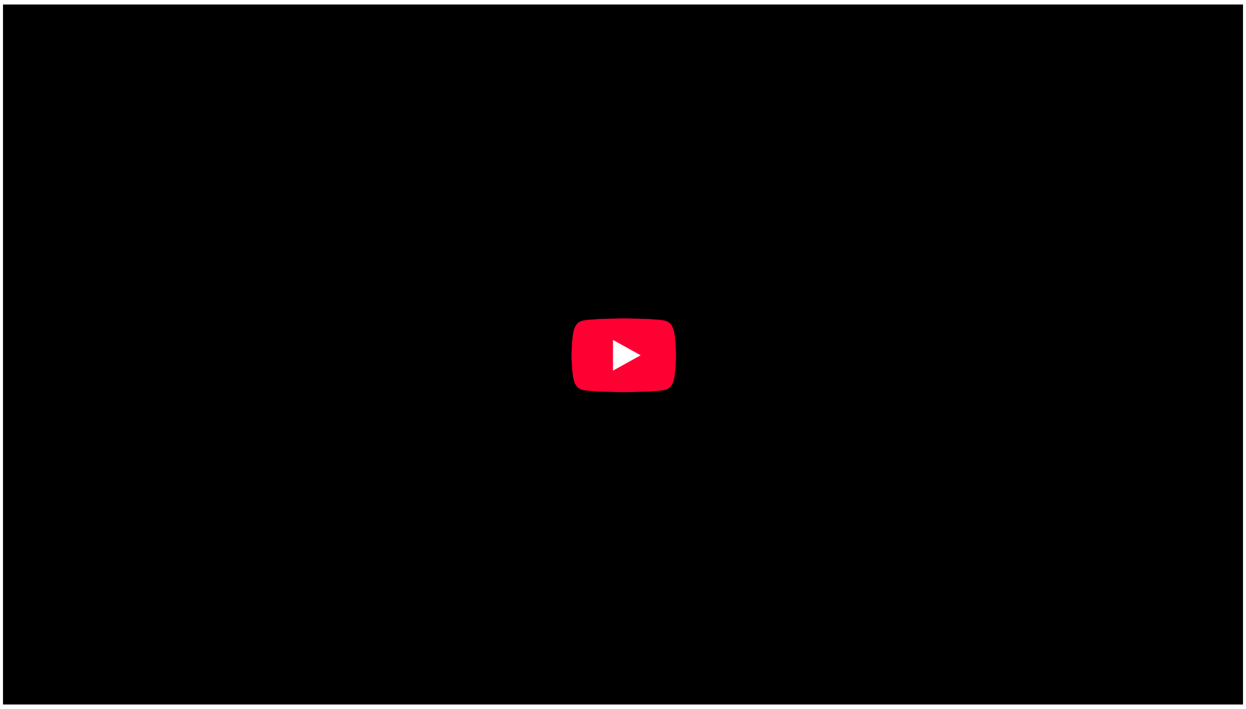
Despite these expanded capabilities, like many AI tools, ChatGPT does have a variety of limitations. Some of its current limitations include:

- Sharing incorrect information: The paid GPT-4 version of ChatGPT has access to the internet, while GPT-3.5 (the free version) does not and can only access information prior to 2021. Since ChatGPT is a predictive text model, it can at times make up information, especially if the information is not easily accessible. This makes ChatGPT prone to making factual mistakes, often confusing similar information, and even making up citations when asked to produce them.
- Generating personal reflections: ChatGPT can generate responses to questions and prompts, but it is still an AI bot; thus, it is unable to respond to a prompt that requires a student's personal experience or reflections.
- Producing non-text based responses: Despite the availability of other image-based AI generators such as Dall-E 2 and MidJourney, ChatGPT responses are strictly text-based. Therefore, any response produced can only offer text. With its most recent updates, ChatGPT can, however, accept image input prompts to generate captions and produce image analyses.

An awareness of ChatGPT's capabilities and limitations can help instructors talk openly with their students about the potential role of the tool in the course and what is expected in terms of student engagement with AI tools. Additionally, knowing what ChatGPT can and cannot do can also help instructors make decisions around course and assignment design. The following section offers several approaches for instructors to consider when developing their own approaches to ChatGPT in their own courses. Instructors can explore OpenAI's documentation Educator considerations for ChatGPT as they consider their own approach to teaching and learning with ChatGPT.

ChatGPT in the Classroom

There is no one correct way to navigate the use of ChatGPT in the classroom. The role it plays (or doesn't) in a course will depend on several factors including course objectives and goals, disciplinary skills that students need to practice, and instructors' own comfort level with engaging with the tool. No matter how instructors respond to the evolution of AI tools, it's important to have explicit expectations, and provide opportunities for clear conversations with students about those expectations. Read on to learn more about several possible approaches and considerations to help instructors start navigating, leveraging, and possibly even engaging ChatGPT (and future AI tools) in their course.



View recording of the "Teaching and ChatGPT" forum held on February 13, 2023. Columbia contributors shared resources and facilitated an informal conversation about AI tools, specifically ChatGPT, and its implications in the classroom.

Develop Course Policies that Include Digital Transparency

It is important to be explicit with students about the expectations around the usage of ChatGPT and other AI tools in your course. For example, ChatGPT is capable of reading a student's essay and providing meaningful feedback that can then be used by the student to make edits. As the instructor, be clear about these expectations: can students use the tool for feedback on their own writing? If so, how should they disclose their use? As with all course policies, especially those around academic integrity, it is essential for instructors to be explicit and transparent with their expectations, and to have frank conversations with their students. Some colleagues are collecting and maintaining an open-source repository of sample digital transparency language and policies from higher education. While not affiliated with Columbia, this collection can offer some insight into the different approaches institutions and individual instructors are taking when it comes to addressing AI tools in the classroom.

Since ChatGPT's introduction, there has been a parallel rise in tools claiming accurate detection of AI-generated work. As with any form of detection software, there are risks of misidentification, which can have consequences in the classroom. These products are best used with careful consideration, and as one of many ways to work with students. It is also important to include the use of these tools in any discussion with students around course policies, making clear why and how such services may be used in the course.

When developing usage policies in a course, instructors might also consider partnering with their students to develop the policies. This partnership can create opportunities for instructors and students to talk in detail about the evolution of particular tools, their potential benefits in specific disciplines, and their limitations. Instructors should be explicit about the course objectives and how the use of these tools might interfere with students' learning and their achievement of particular learning goals.

Scaffold Activities and Assignments

Regardless of AI tool innovation or evolution, one important approach for instructors is to leverage scaffolded activities and assignments. Scaffolding is the process of breaking down a larger assignment into subtasks, which create opportunities for students to check-in and receive feedback. At the same time, scaffolding can help instructors become more familiar with students' work as the semester progresses. This cyclical process of feedback and revision makes the use of tools like ChatGPT challenging and perhaps even unlikely, as students will provide drafts incrementally, and engage in the process of drafting and revision. More importantly, though, this breaking down of a large project into incremental parts helps students to more deeply engage with the different skills and component parts, while also creating valuable time for feedback and reflection throughout the process. For ideas on how to scaffold student work, view the recording of "Using AI writing tools in your scientific writing process" (May 2, 2023) in which Tim Requarth shares how he uses AI tools to streamline and support the writing process.

Design Authentic Assessments for Learning

Authentic assessments centered on student learning can help instructors and learners make intentional choices about the integration of AI tools into the writing process. These kinds of authentic assessments ask students to apply the course concepts they have learned to a "real world" situation or problem. In doing so, authentic assessments can enhance student learning by engaging students in "doing" a particular subject and practicing specific disciplinary skills that will help prepare them for their professional lives outside the classroom. Additionally, these kinds of assignments engage students in higher order thinking and require that they grapple with real world problems and challenges. Designing authentic assessments can ask students to draw from and engage with specific course materials, explore their local community, make connections between course concepts, or also ask students to incorporate their personal experiences or reflections. In this way, authentic assessments can help prevent the use of tools like ChatGPT in that their very design and objectives are rooted within specific course concepts, while also asking students to infuse their own experiences and reflections.

Incorporate AI Tools into Assignment Design

For some courses, depending on the goals and objectives, instructors might consider ways to incorporate AI tools in their assignment design; in doing so, instructors can provide students with opportunities to practice and foster the digital literacy skills they will need for the future. These kinds of creative assignments might ask students to produce AI-written texts as a way to develop awareness of voice, authorship, and accuracy. Additionally, students could apply a rubric and offer feedback on AI-produced texts to build deeper awareness of a course prompt. Lastly, in some instances, ChatGPT might be called upon as a learning support tool, where students ask for feedback on their own texts, have readings summarized, create personalized study materials, or brainstorm for ideas. No matter the assignment design approach taken, instructors should offer opportunities to discuss the assignment with students, asking them to reflect on the experience and analyze their engagement with the tool.

Conclusion

Perhaps unsurprisingly, higher education, like all industries, will continue to feel the impacts of technological evolution and growth. As such, and like they always have, classrooms will continue to remain flexible and responsive to this evolution. With digital innovation and developments, the capabilities (and limitations) of today's AI tools, including ChatGPT, will shift and evolve. For that reason, trying to completely ignore or shut out these tools, or even adopting an approach of complete disengagement, will not serve instructors and their students in the long-term. Instead, instructors have an opportunity to rethink and focus on the elements of their course over which they have the most control, including transparent course policies, explicit communication, partnerships with students, and course and assignment design. Leveraging these aspects of teaching and learning can better serve instructors and their students no matter the digital innovations of the future.

The CTL is here to help!

Looking to develop scaffolded authentic assignments for your course? Want to know more about designing assignments that incorporate AI Tools? Have questions about digital transparency in your course policies? The CTL is here to help – email CTLFaculty@columbia.edu to schedule a 1-1 consultation!

Resources and References

Columbia CTL Related Resources

- Designing Assignments for Learning
- Feedback for Learning
- Getting Started with Creative Assignments
- Learner Perspectives on AI Tools: Digital Literacy, Academic Integrity, and Student Engagement
- Metacognition

Additional Related Resources

- Barnard Center for Engaged Pedagogy (2023). Generative AI & the College Classroom.
- Cornell University Center for Teaching Innovation (2023). Promoting Academic Integrity in Your Course.
- Digital Futures Institute, Teachers College, Columbia University (2023). ChatGPT and Other Artificial Intelligence (AI) in the Classroom.
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- Penn Center for Teaching and Learning (2023). ChatGPT and Its Implications for Your Teaching.
- University of Central Florida Faculty Center. (2023). Artificial Intelligence Writing.
- Yale Poorvu Center for Teaching and Learning (2023). AI Guidance.

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McMurtrie, Beth. (March 2023). ChatGPT is already upending campus practices. Colleges are rushing to respond. *Chronicle of Higher Ed.*

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OpenAI. (November 2022). ChatGPT: Optimizing language models for dialogue. *OpenAI.*

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OpenAI. Educator considerations for ChatGPT. https://platform.openai.com/docs/chatgpt-education

CTL resources and technology for you.

Overview of all CTL Resources and Technology